

VR-Sets. Preprint.

Part III. The ZF Axioms as Closure Theorems

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0. Introduction

This part of the preprint shows that the eight standard ZF axioms (together with the axiom of choice AC) are, in VR-Sets, not postulates of existence but closure theorems of the operational universe relative to specific describable constructions. Two of them — extensionality and the existence of the empty set — have already been proved as Lemmas 1 and 2 of Part II and are only recalled here. The remaining seven assertions (pairing, union, power, infinity, replacement, foundation, choice) are derived by a uniform pattern on the basis of the closure principle fixed in Part II.

Proof pattern. For each axiom:

(1) the classical ZF formulation is given; (2) the operational functionality F corresponding to the asserted set is specified; (3) the operational describability of F is verified — that the functionality gives a definite response to every query; (4) by the closure principle it is concluded that F is a set.

Summary of results of Part III. The eight ZF axioms together with AC receive the following status in VR-Sets:

Axiom	Status in VR-Sets
Extensionality	Lemma 1 (consequence of Definition $\equiv$)
Empty set	Lemma 2 (consequence of Definition 2)
Pairing	Closure theorem (trivially)
Union	Closure theorem (with a remark on $\equiv$)
Power	Closure theorem; $\wp(\omega)$ is countable (divergence)
Infinity	Theorem via operational infinity
Replacement	A single theorem in place of a schema (simplification)
Foundation	Mode-dependent: theorem in ZFC, false in ZFA
Choice	Theorem on the countable operational universe

Terminological reminder. All terms — operational set, membership functionality, operational depth, operational identity $\equiv$, the closure principle — were introduced in Part II and the Foundation Notes and are used here without redefinition. The sign $\equiv$ denotes the

operational identity of sets; the sign of equality = does not appear in this part as an independent symbol, except in classical ZF formulations.

Three critical points. Before the detailed proofs we highlight three places where the results of this part are especially important for what follows:

- $\wp(\omega)$ in VR-Sets is countable. The sharpest divergence from classical ZF; it shows up already here, before the separate chapter on uncountability (Part VI).
- The replacement schema collapses into a single theorem. A structural simplification: the closure principle covers what in ZF requires an axiom schema.
- Foundation is a mode-dependent property. The distinction between the ZFC-mode and the ZFA-mode is defined precisely through the satisfaction or non-satisfaction of foundation. In Part IV this becomes a technical definition.

III.1. Extensionality

Classical ZF formulation

$$\forall A \forall B (\forall x (x \in A \leftrightarrow x \in B) \rightarrow A = B)$$

Status in VR-Sets

Assertion (Lemma 1 of Part II). If sets A and B have the same elements (up to operational identity $\equiv$), then $A \equiv B$.

Remark. This is not a closure theorem: extensionality is built into the very definition of operational identity (Definition 4). The condition "for every element $x \in A$ there exists $y \in B$ such that $x \equiv y$, and conversely" is precisely the condition of coincidence of functionalities, i.e. $\equiv$ by definition. The statement "sets with the same elements are distinct" contradicts the definition of identity and is inexpressible in VR-Sets.

Relation to ZF. In ZF extensionality is a separate axiom because in the classical approach equality is given independently of membership. In VR-Sets equality (more precisely, operational identity) is *defined through* membership, and hence extensionality is a tautology of the definition. This is the first (and gentlest) of the places where the apparatus of VR-Sets proves more compact: a single decision at the level of definitions removes the need for a separate axiom.

III.2. Empty set

Classical ZF formulation

$$\exists A \forall x (x \notin A)$$

Status in VR-Sets

Assertion (Lemma 2 of Part II). There exists exactly one set with an empty functionality; it is denoted $\emptyset$ and is given as a primitive by Definition 2.

Remark. The existence of $\emptyset$ is not a closure theorem but a primary primitive of the ontology: $\emptyset$ is the sole entity for which ontological primacy is asserted ("only $\emptyset$ is"). Uniqueness was proved in Lemma 2: if two functionalities both reveal nothing, the condition of coincidence of functionalities in Definition 4 holds vacuously on both sides.

Relation to ZF. In ZF the existence of the empty set is postulated by a separate axiom and justified by the separation schema. In VR-Sets $\emptyset$ is the sole ontological primitive, and there is no need to postulate it: $\emptyset$ is given by the construction of the system. The existence of the empty set in VR-Sets is neither a theorem nor an axiom but an ontological decision underlying the entire foundation.

III.3. Pairing

Classical ZF formulation

$$\forall a \forall b \exists A \forall x (x \in A \leftrightarrow x \equiv a \vee x \equiv b)$$

Theorem (pairing)

For any sets a and b there exists a set, denoted $\{a, b\}$, whose functionality reveals exactly a and b (when $a \equiv b$ — a single element).

Proof

Functionality. Define $F_{\{a,b\}}$ as follows: on the first query the functionality reveals a ; on the second — reveals b ; on all subsequent queries reveals nothing. If $a \equiv b$, the second step amounts to a repeated presentation of the same element, and by extensionality (Lemma 1) the functionality reveals a single element.

Describability. On every concrete query the response is defined: 1st query $\rightarrow a$; 2nd query $\rightarrow b$; n -th query for $n \geq 3 \rightarrow$ exhaustion. No indeterminacy or contradictory responses arise.

Conclusion. By the closure principle, $F_{\{a,b\}}$ is a set. It is denoted $\{a, b\}$. ■

Remark: ordered pairs

The ordered pair $\langle a, b \rangle$ is **not** introduced as an object of VR-Sets. This is consistent with the position of VR-Numbers, where pairs are excluded from the ontology: "only $\emptyset$ is, all else is doing" means that new entities are not produced by "binding" already-existing ones, only by describable functionalities.

If in the sequel (for instance, in relations and functions) order should be required, it is realised not by introducing pair-objects but through *the order of revelation of elements within a functionality*. Order becomes an internal property of the description of a functionality, not a separate structure. Technically this means that a function $f : A \rightarrow B$ in VR-Sets is an operational set whose functionality, on the query "what does f yield at a ?", returns the corresponding element of B , without the mediation of a set of pairs $\langle a, f(a) \rangle$.

III.4. Union

Classical ZF formulation

$$\forall A \exists B \forall x (x \in B \leftrightarrow \exists C (C \in A \wedge x \in C))$$

Theorem (union)

For any set A there exists a set, denoted $\bigcup A$, whose functionality reveals exactly all elements of elements of A (without repetitions up to $\equiv$).

Proof

Functionality. Define $F_{\bigcup A}$ as a nested procedure: for each element C revealed by the functionality A , reveal in turn all elements x revealed by the functionality C ; elements already previously revealed (up to $\equiv$) are skipped.

Describability. The functionality A is describable (A is the given set); for each revealed C the functionality C is describable (C is a set, an element of A); the composition of describable functionalities is a describable procedure. On every concrete query $\bigcup A$ gives a definite response: either a new element, or exhaustion (when all elements of all $C \in A$ have already been revealed).

Conclusion. By the closure principle, $F_{\bigcup A}$ is a set. It is denoted $\bigcup A$.

■

Subtlety: decidability of $\equiv$ when skipping repetitions

The formulation of the functionality used the condition "skipping repetitions up to $\equiv$." In general, operational identity $\equiv$ is not algorithmically decidable: for arbitrary infinite sets A and B the question " $A \equiv B$?" reduces to a coinductive check of coincidence of functionalities and in the worst case fails to terminate in a finite number of steps.

This does **not** render the functionality $\bigcup A$ impossible: its operational description is correct — at each step the decision "yield an element or skip" depends on the true answer to the question "does it coincide with one of those already yielded?" However, realising this functionality by an *effective algorithm* is not always possible. This distinction is consistent with the general position of VR-Sets: the theory is operational but not necessarily effective throughout. Decidability of $\equiv$ is a separate question for separate classes of sets (for finite sets with finite operational depth $\equiv$ is decidable; for VR numbers $\equiv$ is decidable by construction; in general — not).

An alternative formulation without appeal to $\equiv$: $F_{\bigcup A}$ reveals all elements of all $C \in A$ *with possible repetitions*, and extensionality (Lemma 1) guarantees that the resulting set coincides with the set without repetitions. In this form describability does not depend on the decidability of $\equiv$, and the proof becomes more rigorous. In the sequel we take this as the canonical formulation.

III.5. Power

Classical ZF formulation

$$\forall A \exists B \forall x (x \in B \leftrightarrow x \subseteq A)$$

Theorem (power)

For any set A there exists a set, denoted $\wp(A)$, whose functionality reveals exactly all subsets of A — that is, all operationally describable functionalities whose revealed elements belong to A .

Proof

Functionality. Define $F_{\wp(A)}$ as a procedure revealing in turn all subsets of A .

Case of finite A . Let A have operational depth n . There are 2^n subsets; each is given by a functionality "reveal such-and-such elements of A , the others not at all," which is trivially describable. The enumeration of 2^n subsets is finite and describable.

Case of infinite A (for instance $A = \omega$). A subset of ω is any operationally describable functionality revealing some elements of ω . The set of such describable functionalities is **countable**: each functionality is given by a finite description (an algorithm of finite length) over a finite alphabet, and the set of finite strings over a finite alphabet is countable. The enumeration of these functionalities is describable (for instance, by increasing length of description and lexicographically within a single length).

Describability. In both cases, on every concrete query $F_{\wp(A)}$ gives a definite response: the n -th subset of A by the enumeration, or exhaustion (for finite A).

Conclusion. By the closure principle, $F_{\wp(A)}$ is a set. It is denoted $\wp(A)$.

■

Important consequence: $\wp(\omega)$ in VR-Sets is countable

From the proof it follows that $\wp(\omega)$ in VR-Sets is a countable set. This is a sharp divergence from classical ZF, where $\wp(\omega)$ is uncountable by Cantor's diagonal argument. The divergence calls for a direct comment.

A clarification on the status of $\wp(\omega)$. The proof above establishes $\wp(\omega)$ as a set by the closure principle: the description "the operability revealing in turn all describable subsets of A " specifies a functionality, and by closure this functionality is a set. This is "existence as an object." It is a separate question whether the same enumeration is realisable as an effective operational procedure — whether there exists an algorithm that, on the n -th query, returns a correct description of the n -th subset. The answer is no: such an algorithm would have to decide, for an arbitrary description, whether it specifies a correct functionality, which is equivalent to the halting problem. $\wp(\omega)$ is therefore countable as an object but not enumerable as a procedure. This distinction — countable from without vs. enumerable from within — is developed in detail in Part VI §VI.2 and §VI.3, where it is shown to be precisely what removes Cantor's diagonal in the operational ontology.

Where Cantor's diagonal fails to apply. The classical diagonal argument constructs, from an enumeration of subsets $S_0, S_1, S_2, \dots$, a set $D = \{n : n \notin S_n\}$ differing from every S_n . For this argument the enumeration of the S_n must be complete, and the definition of D must specify an operational functionality.

In VR-Sets a complete enumeration of all describable subsets of ω is **operationally unattainable** — it is equivalent to solving the halting problem: to enumerate all algorithms correctly specifying a subset of ω one would have to separate terminating functionalities from non-terminating ones, which is undecidable. Accordingly, the diagonal D , defined with respect to "all describable subsets," does not specify a describable functionality — it rests on an operational procedure which does not exist. By the closure principle D is not a set.

The diagonal is thus removed *not by forbidding* it as a proof, but by the fact that its premise (a complete enumeration of describable subsets) is operationally unrealisable.

Connection with computable analysis. An analogous situation is well known for computable real numbers: the set of computable reals is countable "from without" (from the standpoint of classical mathematics) but not enumerable "from within" — there is no algorithm enumerating all computable reals. $\wp(\omega)$ in VR-Sets stands in the same position: countable as an object, not enumerable as a functionality.

A detailed discussion of the consequences appears in Part VI. Here let us fix: the power set exists as a closure theorem, but **$\wp(\omega)$ in VR-Sets and $\wp(\omega)$ in classical ZF are not one and the same object**. The difference is most visible precisely at this axiom.

III.6. Infinity

Classical ZF formulation

$$\exists A (\emptyset \in A \wedge \forall x (x \in A \rightarrow x \cup \{x\} \in A))$$

Theorem (infinity)

There exists a set, denoted ω , whose functionality reveals in turn $\emptyset, t(\emptyset), t(t(\emptyset)), \dots$ without exhaustion and without cycling.

Proof

Functionality. Define F_ω : on the n -th query (for $n \geq 0$) the functionality reveals the n -th von Neumann ordinal O_n , where $O_0 = \emptyset$ and $O_{n+1} = t(O_n)$ (definitions from Part I of VR).

Describability. Each step of revelation is defined: the operator t (successor) is total on the objects of VR (axiom A3); by induction (axiom A4) the functionality is defined for every n . On every concrete query F_ω gives a definite response — a specific object O_n . The functionality does not terminate (is not exhausted at any finite step) and is non-cyclic ($t(x) \neq x$ in VR — Theorem 5 of Part I).

Conclusion. By the closure principle, F_ω is a set. It is denoted ω .

■

Remark: operational infinity

This statement lifts to the set level the principle of *operational infinity* from VR-Numbers (Part VI.3). In classical ZF the axiom of infinity postulates the *existence of a completed totality* of all natural numbers; finite means cannot derive its existence, so it is required as an axiom.

In VR-Sets no "completed totality" is asserted. ω is an **unboundedly applicable operation** enumerating O_n , and by the closure principle an unboundedly applicable describable operation is already a set. The divergence with classical ZF at this point is methodological: where ZF postulates, VR-Sets discovers a consequence of the operational ontology.

ω has *non-finite* operational depth (unfolding to $\emptyset$ does not terminate in any finite number of steps), but is **non-cyclic**: the sequence $O_0, O_1, O_2, \dots$ is non-repeating. This distinction "infinite by enumeration" vs. "cyclic" first becomes operative at ω .

III.7. Replacement

Classical ZF formulation (axiom schema)

For every formula $\varphi(x, y)$ specifying a functional relation ($\forall x \exists! y \varphi(x, y)$) and for every set A there exists a set $B = \{ y : \exists x \in A, \varphi(x, y) \}$.

In ZF this is an **axiom schema** — one axiom for each formula φ of the language of ZF.

Theorem (replacement)

For any set A and any operationally describable procedure F defined on the elements of A , there exists a set, denoted $\text{image}(F, A)$, whose functionality reveals exactly $F(x)$ for every $x \in A$.

Proof

Functionality. Define F_{image} : for each element x revealed by the functionality A , reveal $F(x)$.

Describability. The functionality A is describable. The procedure F is describable by assumption (this is part of the hypotheses of the theorem). The composition of describable functionalities is a describable functionality. On every query F_{image} gives a definite response: for the next revealed $x \in A$ the result is a specific object $F(x)$, or exhaustion if A is exhausted.

Conclusion. By the closure principle, F_{image} is a set. It is denoted $\text{image}(F, A)$. ■

Structural simplification: schema $\rightarrow$ single theorem

In classical ZF replacement is a schema: one axiom for each formula φ . This is bound up with the fact that in ZF a "relation" is a syntactic object (a formula of the language), and for each formula a separate existence assertion is required.

In VR-Sets an "operationally describable procedure F " is a single ontological notion encompassing all ways of specifying functional relations (including all formulas of the

language of ZF translated into operational descriptions). Therefore **a single closure theorem with respect to describable transformations replaces the entire schema.**

This is a pattern typical of VR-Sets: what in ZF is dispersed over an infinite series of syntactic cases is, in VR-Sets, gathered into a single substantive assertion resting on the ontology (the closure principle), not on syntax.

Comparison with Fraenkel's replacement. Replacement was added to ZF by Fraenkel and Skolem in order to construct sets such as $\{ \omega, \omega+1, \omega+2, \dots \}$, unreachable by Zermelo's means. In VR-Sets such sets are obtained directly through the closure principle: the procedure "apply t k times to ω " is describable, and so the image of this procedure on ω is a set. No additional postulates are needed.

III.8. Foundation (Regularity)

Classical ZF formulation

$$\forall A (A \neq \emptyset \rightarrow \exists x \in A (x \cap A = \emptyset))$$

Equivalent formulation

There exist no infinitely descending chains of membership: there is no sequence $x_0 \ni x_1 \ni x_2 \ni \dots$ without termination.

Status in VR-Sets: mode-dependent

This is the first ZF axiom whose status in VR-Sets **depends on the choice of mode** (ZFC-mode vs. ZFA-mode). Unlike the preceding axioms — closure theorems holding unconditionally — foundation turns out to be a *characterising property* of one of the two modes.

Assertion 1 (foundation in the ZFC-mode)

In the ZFC-mode of VR-Sets, foundation holds: there exist no infinitely descending chains of membership.

Proof

In the ZFC-mode, self-references of a label to its own functionality are syntactically forbidden. We show by induction on operational depth (Lemma 3 of Part II) that every set in the ZFC-mode has finite depth or "infinite-but-foundationally-grounded-by- ω -enumeration" depth.

Base. $\emptyset$ has depth 0. There is no membership chain from $\emptyset$ ($\emptyset$ contains no elements).

Step. Suppose the assertion holds for all sets of depth $< n$. Let A have depth n . Any chain $A \ni x_1 \ni x_2 \ni \dots$ begins with $x_1 \in A$; the depth of x_1 is strictly less than n (by the definition of operational depth); by the inductive hypothesis the chain from x_1 terminates at $\emptyset$. Hence the original chain terminates.

Case of ω -depth. For sets such as ω whose depth "runs to infinity" (and is non-cyclic), a chain from ω has the form $\omega \ni O_n \ni O_{n-1} \ni \dots \ni O_0 = \emptyset$. Any specific choice of an

element $x_1 \in \omega$ is a specific O_n with finite n , and the subsequent chain terminates at $\emptyset$. An infinitely descending chain from ω does not arise.

■

Assertion 2 (failure of foundation in the ZFA-mode)

In the ZFA-mode foundation **does not** hold. There exist sets giving infinitely descending chains of membership.

Example

In the ZFA-mode the description of a set A is admissible through a label $A :=$ the functionality revealing one element — A itself. Formally: the functionality "on a query reveal the set denoted by the same label as the current one." The chain $A \ni A \ni A \ni \dots$ does not terminate. This set is known as a Quine atom in Aczel (1988).

Here it is important: A does **not contain itself** in the sense of physical containment. According to the ontology of VR-Sets (Foundation Notes §2), membership is a *reference*, not a container. $A \in A$ means that the functionality A returns A on a query — a specific definite response, not a structural self-embedding. There is no paradox; there is only a failure of foundation.

Conclusion

Foundation is not a universal theorem of VR-Sets but a **characterising property of the ZFC-mode**. The two modes differ precisely on this:

- **ZFC-mode** — foundation holds (the theorem above); self-references are syntactically forbidden; the operational depth of every set is defined.
- **ZFA-mode** — foundation fails; self-references are admitted; the operational depth of certain sets is undefined (cyclic functionalities).

In Part IV this will give a technical definition of the distinction between the two modes: ZFC-mode = ZFA-mode + foundation (when the restriction on self-references of labels is properly formulated). Conjecture: the ZFC-mode is equivalent to classical ZFC, the ZFA-mode to Aczel (AFA).

III.9. Choice

Classical ZF formulation (AC)

For every set of pairwise disjoint non-empty sets there exists a set containing exactly one element from each.

Status in VR-Sets: theorem on the countable universe

The position on AC is fixed in the Foundation Notes (§6.3) and is unfolded here into a proof. Briefly:

- **DC (dependent choice)** is built into the operational ontology: every operation is carried out concretely, and a sequence of operations is built step by step. DC is not an axiom but a property of operationality itself.
- **Full AC** is a theorem on the countable operational universe (for countable sets, AC reduces to DC).
- **The Banach–Tarski paradox** is not reproduced — not because AC is "operationally interpreted" but because there is no uncountable classical $\mathbb{R}$ in VR-Sets. Disputed applications of AC take place outside the operational universe.

Theorem (countable choice)

For every countable family $\{A_i\}_{i \in \omega}$ of pairwise disjoint non-empty sets there exists a set, denoted $\text{choice}(\{A_i\})$, containing exactly one element from each A_i .

Proof

Functionality. Define F_{choice} : on the i -th query the functionality reveals the first element of A_i — that is, the element revealed by the functionality A_i on its first query.

Describability. The countable family is an operational set whose functionality reveals $A_0, A_1, A_2, \dots$. Each A_i is an operational set with a non-empty functionality, so its first query gives a definite element. The composition of these describable procedures specifies a describable functionality F_{choice} : for every $i \in \omega$ a specific response is defined.

Conclusion. By the closure principle, F_{choice} is a set.

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On full AC and the countability of the universe

The theorem just proved suffices for the entire operational universe of VR-Sets: the whole universe is countable (Foundation Notes §6.2), and so **any** family of non-empty pairwise disjoint sets in VR-Sets is countable. There are no uncountable families in VR-Sets. Accordingly, AC in full form is equivalent to countable AC, which is a theorem.

A detailed discussion appears in Part VI. Here let us fix: **AC is a theorem, not an axiom.** The Banach–Tarski paradox is absent not because AC is weakened, but because the objects on which it is built (uncountable subsets of $\mathbb{R}$, non-Lebesgue-measurable sets) do not exist in the operational ontology.

Subtlety: the operationality of "first element"

The formulation "reveal the first element of A_i " requires that the functionality A_i have a fixed "first" query. By Definition 1 of Part II the order of revelation of elements is an artefact of observation and does not enter into the identity of a set. However, for each concrete *presentation* of the functionality A_i its first query yields a definite element. If there are several presentations and the order of revelation differs, choice may return different elements — but these are different functionalities choice, not an inconsistent definition.

Formally: with a fixed order of revelation of each A_i , the function choice is uniquely defined. The operational theorem says that *such* a functionality is a set. It does not presuppose that

there is a uniquely correct choice (which depends on the order of presentation); only that any consistent choice specifies a set. This is precisely the content of classical AC: existence of a choice function, not its uniqueness. The operational reformulation preserves this classical content exactly.

Conclusion

Part III shows that the ZF axioms in VR-Sets are not postulates but theorems. Of the nine assertions considered:

- **Two** (extensionality, empty set) are built into the definitions and lemmas of Part II and do not require a separate proof.
- **Six** (pairing, union, power, infinity, replacement, choice) are closure theorems of the operational universe with respect to specific describable functionalities. The closure principle gives them a uniform proof.
- **One** (foundation) is a mode-dependent property distinguishing the ZFC-mode from the ZFA-mode.

Substantive divergences from classical ZF brought out in this part:

- (1) $\wp(\omega)$ in VR-Sets is countable. Cantor's diagonal argument in the operational ontology produces no new subset, because its premise (a complete enumeration of describable functionalities) is operationally unrealisable — it is equivalent to solving the halting problem.
- (2) *The replacement schema reduces to a single theorem.* The closure principle with respect to describable transformations covers the entire series of cases that, in ZF, require a separate axiom for each formula.
- (3) *Foundation is not a universal property but a distinguishing one.* The ZFC-mode and the ZFA-mode in VR-Sets are two ontologically legitimate configurations; the distinction between them is a syntactic restriction on self-references of labels, technically manifesting as the satisfaction or non-satisfaction of foundation.
- (4) *AC is a theorem, not an axiom.* On the countable operational universe AC reduces to DC, which is built into operationality. The Banach–Tarski paradox is not reproduced, not because of a weakening of AC but because of the absence of the classical $\mathbb{R}$ objects on which it is built.

What follows. Part IV uses foundation as the technical definition of the distinction between the ZFC-mode and the ZFA-mode, formulates this distinction syntactically, and discusses the conjecture of equivalence with classical ZFC and with Aczel. Part V specifies the role of VR numbers in VR-Sets: O_n are specific operational sets with finite depth, landing in the ZFC-mode as a theorem. Part VI develops the countability of the operational universe and discusses Cantor's diagonal, AC, and the absence of uncountability paradoxes in detail.